

PRACTICAL: 01

Aim: Implementation to gather information from any PC's connected to the LAN using who.is, port scanners, network scanning, Angry IP scanners etc.

Requirements: Laptop, who.is, n map, angry ip scanner

Who.is

What is a WHOIS Search?

A **WHOIS search** is a specialized query used to obtain registration information about an object stored in a WHOIS database. This protocol is widely used for querying details about Internet resources such as domain names and IP address blocks. It can also be used for other types of network-related objects. The information retrieved may include the registered owner, administrative contacts, creation and expiration dates, and other vital statistics about the resource.

How WHOIS Works

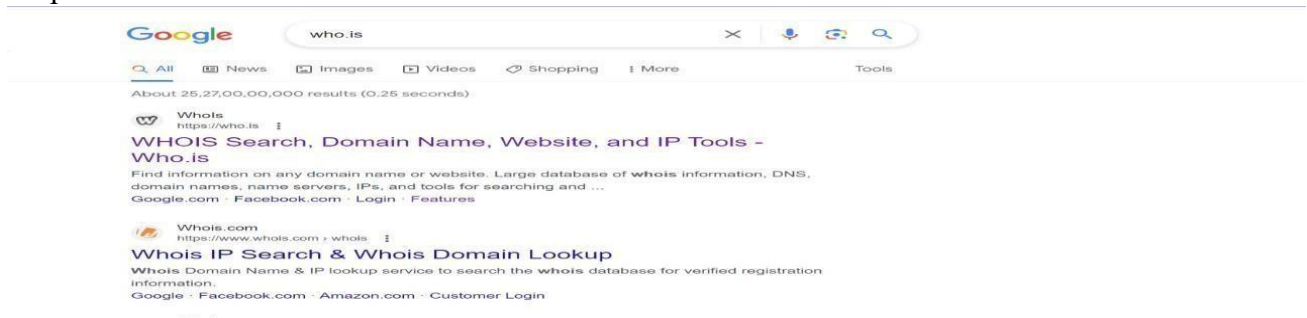
- **Protocol:** WHOIS operates using a query and response protocol based on TCP.
- **Purpose:** It connects to a WHOIS server and retrieves publicly available registration details about a queried domain or IP address.
- **Server Selection:** Most modern WHOIS tools automatically determine the correct server for your query. If they can't, they default to standard servers (such as whois.networksolutions.com for network handles or whois.arin.net for IP addresses).

Common WHOIS Search Examples

- **Domain Lookup:** Retrieve WHOIS details for a domain, such as computersolutions.com, to view ownership and availability.
- **IP Address Lookup:** Obtain the WHOIS record for a specific IP address to identify the owning organization.
- **Search Engine Query:** Inputting "WHOIS domainname.com" in a search engine like Google often provides links directly to WHOIS lookup tools.

Implementation

Step 1: search who.is in web browser



Step 2: open the who.is tool



The screenshot shows the 'Whois Domain Lookup' website. At the top, there is a navigation bar with links: Domains, Hosting, Servers, Email, Security, Whois, Deals. A search bar at the top right contains the text 'Enter Domain or IP' and a 'WHOIS' button. The main heading is 'Whois Domain Lookup' with the subtitle 'Whois search for Domain and IP'. Below this is a large input field with the placeholder text 'Enter Domain Name or IP Address' and a 'SEARCH' button. An example text below the input field reads: 'Example: qq.com, google.co.in, bbc.co.uk, ebay.ca'.

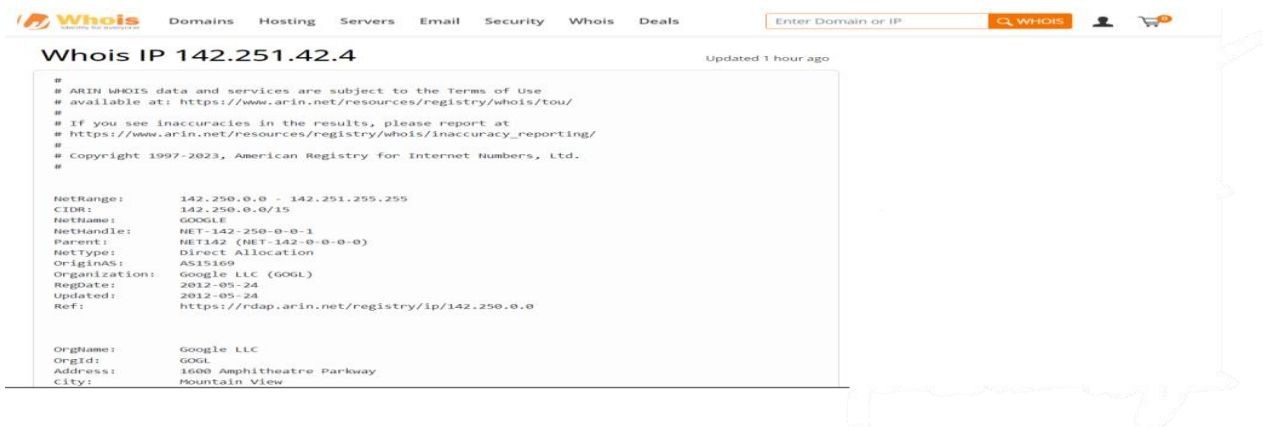
Step 3: Enter the Domain name or IP Address of which devices' information you want to gather



This screenshot shows the same 'Whois Domain Lookup' website, but the input field now contains the IP address '142.251.42.4'. The 'SEARCH' button is still visible, and the example text remains the same.

★ Observation:

Get the information about searched domain name or IP Address



The screenshot shows the results page for the IP address '142.251.42.4'. The page title is 'Whois IP 142.251.42.4' and it says 'Updated 1 hour ago'. The main content area displays the following information:

```
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
# Copyright 1997-2023, American Registry for Internet Numbers, Ltd.
#

NetRange: 142.250.0.0 - 142.251.255.255
CIDR: 142.250.0.0/15
NetName: GOOGLE
NetHandle: NET-142-250-0-0-1
Parent: NET142 (NET-142-0-0-0-0)
NetType: Direct Allocation
OriginAS: AS15169
Organization: Google LLC (GOGL)
RegDate: 2012-05-24
Updated: 2012-05-24
Ref: https://rdap.arin.net/registry/ip/142.250.0.0

OrgName: Google LLC
OrgId: GOGL
Address: 1600 Amphitheatre Parkway
City: Mountain View
```

2. Port Scanners:

Nmap is convenient during penetration testing of networked systems. Nmap provides the network details, and also helps to determine the security flaws present in the system. Nmap is **platform-independent** and runs on popular **operating systems** such as **Linux, Windows** and **Mac**.

Nmap is a useful tool for network scanning and auditing purposes.

- It can search for hosts connected to the Network. ○ It can search for free **ports** on the target host.
- It detects all services running on the host with the help of **operating system**.
- It also detects any **flaws** or **potential vulnerabilities** in networked systems.

It is effortless to work with the Nmap. With the release of a new graphical user interface called **GenMap User**, it performs many tasks such as saving and comparing scan results, scanning the results in a database, and visualize the network system topology graphically, etc. ● Advantages of Nmap:

Nmap has a lot of advantages that make it different from other network scanning tools.

Nmap is open-source and **free** to use.

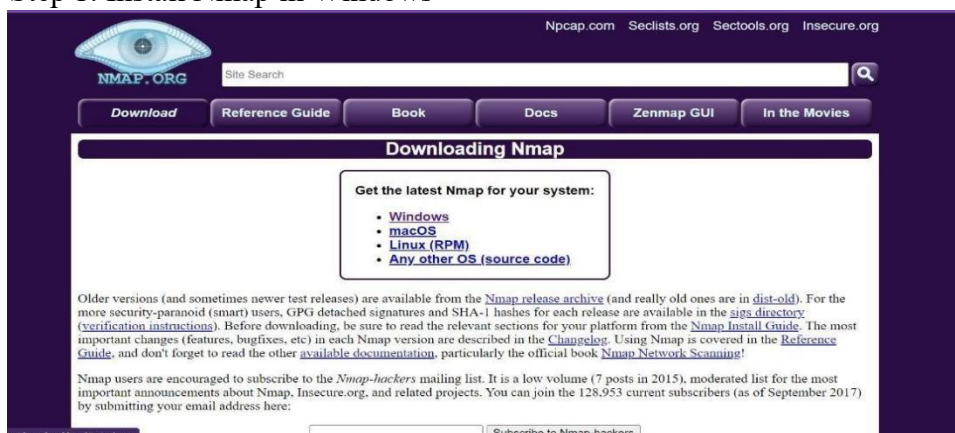
Some other advantages are listed below.

- It is used for auditing network systems as it can detect new servers.
- It will search for **subdomain** and Domain Name System
- With the help of Nmap Scripting Engine (NSE), interaction can be made with the target host. ○ It determines the nature of the service in the host and performs whether the host is a mail service or a web server

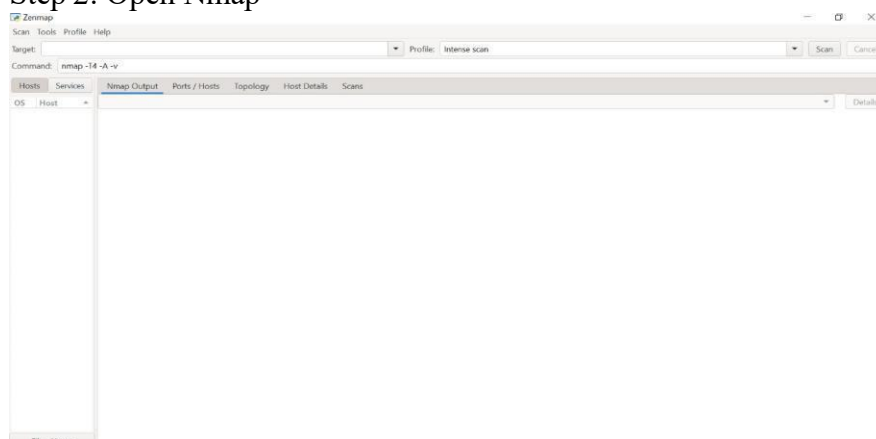
● Port scanning using Nmap:

● **Implementation:**

Step 1: Install Nmap in Windows



Step 2: Open Nmap



Step 3: Open command prompt and give command ipconfig to see local network

Step 4: paste ip address in target select scan type and press scan button

Nmap Input:

```

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : pu.edu

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

    Connection-specific DNS Suffix  . :
    IPv4 Address. . . . . : 192.168.74.8
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.74.209

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\jaysd>|

```

Zenmap

Scan Tools Profile Help

Target: 192.168.29.185
 Profile: Intense scan

Command: nmap -T4 -A -v 192.168.29.185

Hosts Services **Nmap Output** Ports / Hosts Topology Host Details Scans

Nmap Output:

Zenmap

Scan Tools Profile Help

Target: 192.168.29.185
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Command: nmap -T4 -A -v 192.168.29.185

Hosts Services **Nmap Output** Ports / Hosts Topology Host Details Scans

```

nmap -T4 -A -v 192.168.29.185
Nmap scan report for cpe:/o:microsoft:windows_10
PORT      STATE SERVICE          VERSION
135/tcp   open  mrapc            Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
1521/tcp  open  oracle-tns       Oracle TNS listener 11.2.0.2.0 (unauthorized)
8080/tcp  open  http             Oracle XML DB Enterprise Edition httpd
|_ http-server-header: Oracle XML DB/Oracle Database
|_ http-title: 401 Unauthorized
|_ http-auth:
|_ HTTP/1.1 401 Unauthorized\x0D
|_ Basic realm=XDB
|_ http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
Device type: general purpose
Running: Microsoft Windows 10|11
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 10 1607 - 11 23H2
Uptime guess: 9.357 days (since Tue Jul 15 10:55:05 2025)
Network Distance: 0 hops
TCP Sequence Prediction: Difficulty=260 (Good luck!)
IP ID Sequence Generation: Incremental
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb2-time:
|_   date: 2025-07-24T13:59:00
|_   start_date: N/A
|_   smb2-security-mode:
|_     3.1.1:
|_       Message signing enabled but not required

NSE: Script Post-scanning.
Initiating NSE at 19:29
Completed NSE at 19:29, 0.00s elapsed
Initiating NSE at 19:29
Completed NSE at 19:29, 0.00s elapsed
Initiating NSE at 19:29
Completed NSE at 19:29, 0.00s elapsed
Read data files from C:\Program Files (x86)\Nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 33.85 seconds
Raw packets sent: 1016 (45.418KB) | Rcvd: 2045 (87.302KB)

```


Step 5: Open Command prompt Run as administrator
To see the open ports run:

```
Microsoft Windows [Version 10.0.22631.5624]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>nmap solvetic.com
Starting Nmap 7.97 ( https://nmap.org ) at 2025-07-24 19:41 +0530
Stats: 0:00:57 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 62.10% done; ETC: 19:42 (0:00:31 remaining)
Stats: 0:01:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 72.73% done; ETC: 19:43 (0:00:32 remaining)
Stats: 0:01:36 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 72.93% done; ETC: 19:43 (0:00:33 remaining)
Stats: 0:02:41 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 85.50% done; ETC: 19:44 (0:00:26 remaining)
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
Nmap scan report for solvetic.com (178.33.118.246)
Host is up (0.47s latency).
rDNS record for 178.33.118.246: mail.solvetic.com
Not shown: 981 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    filtered smtp
53/tcp    open  domain
80/tcp    open  http
110/tcp   open  pop3
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
143/tcp   open  imap
179/tcp   filtered bgp
443/tcp   open  https
445/tcp   filtered microsoft-ds
465/tcp   open  smtps
554/tcp   open  rtsp
587/tcp   open  submission
646/tcp   filtered ldp
993/tcp   open  imaps
995/tcp   open  pop3s
1723/tcp  open  pptp

Nmap done: 1 IP address (1 host up) scanned in 204.91 seconds

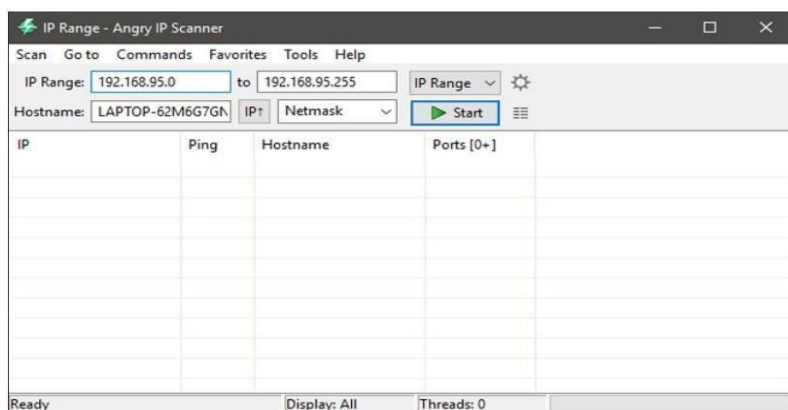
C:\Windows\System32>
```

3. Angry IP Scanner

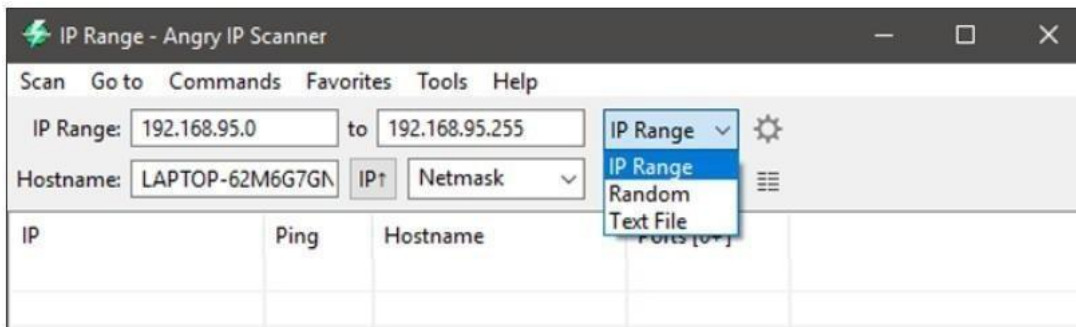
Implementation:



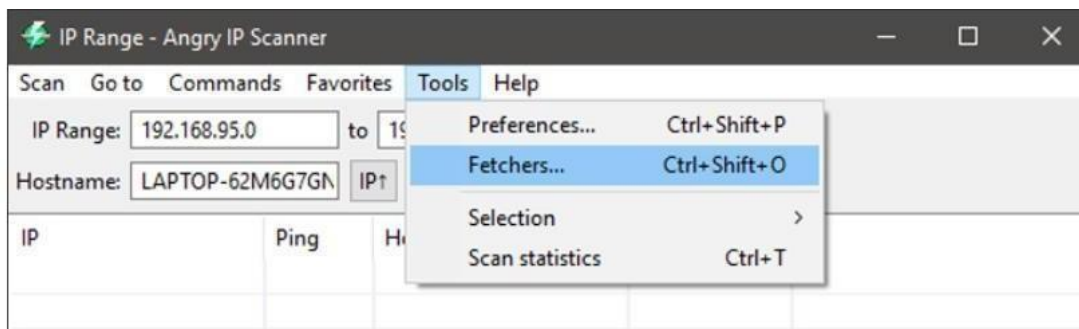
Once installed, open the application by searching for it in the Start Menu. As you can see, the home screen of the application is pretty simple and straightforward. By default, Angry IP scanner will enter your local IP address range and your computer name as the hostname.



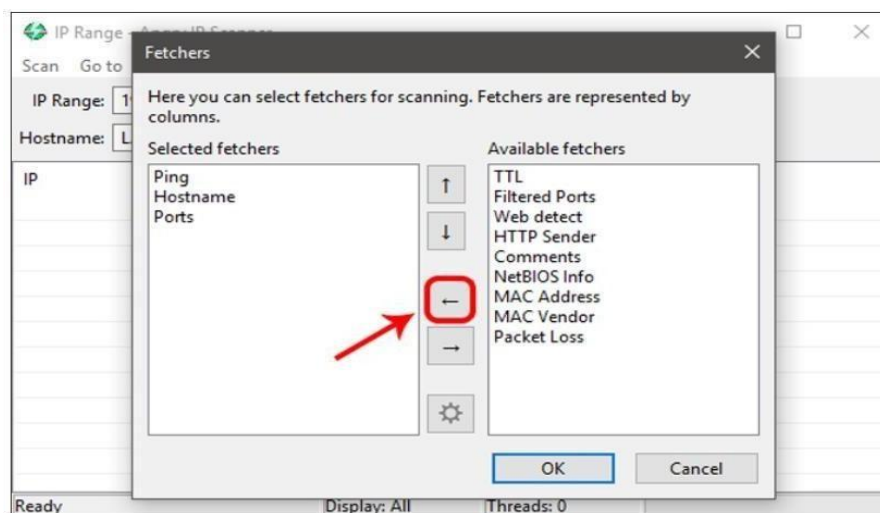
The good thing about Angry IP Scanner is that it lets you scan IP addresses in three different ways. They are, the range you specified, a random IP address or a list of IP addresses from a text file. You can easily select the scan mode from the drop-down menu next to the IP address field.



As you can see from the above image, the Angry IP Scanner will only include default fetchers like Ping, Hostname, and Ports. However, you can add more fetchers to get and see more information about an IP address. To do that, select “Tools > Fetchers.”

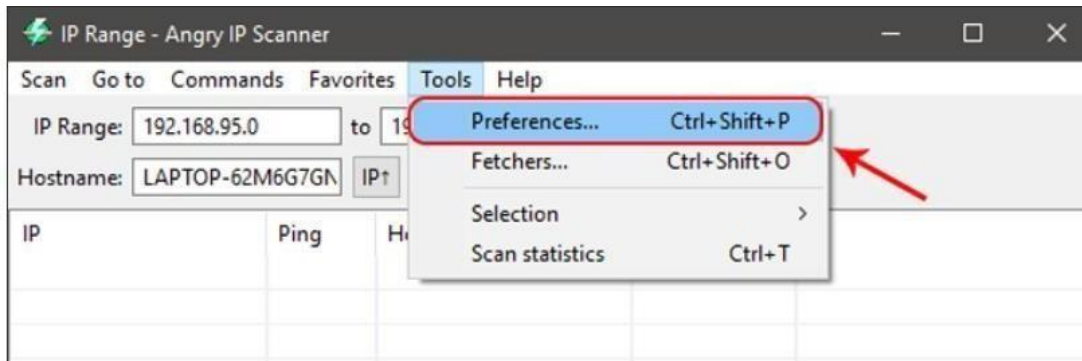


In this window, you will see all the current fetchers on the left pane and all the available fetchers in the right pane. To add a fetcher, select the fetcher on the right pane and then click on the button that looks like “Less than” sign. In my case, I’ve added new fetchers like MAC address, NetBIOS info, filtered ports, and Web detectors.

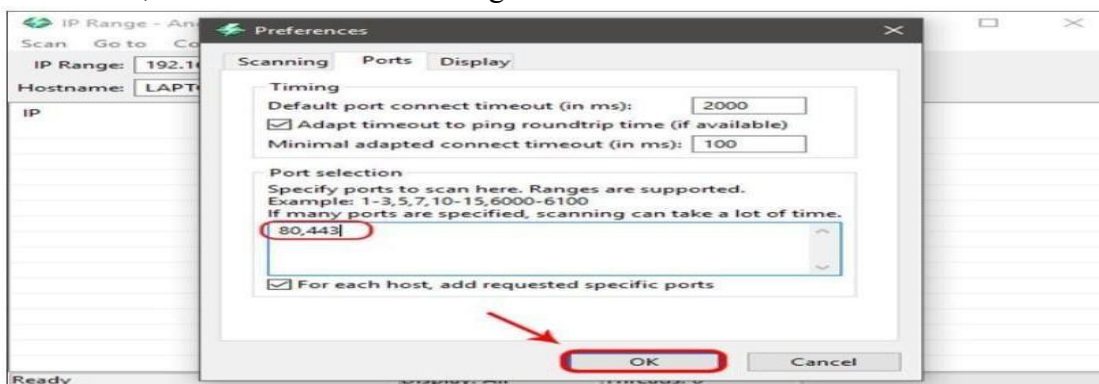


Moreover, Angry IP Scanner will only tell whether the ports are open or not. It will not list the individual ports that are open. So, if you want to do a port scan, then you need to

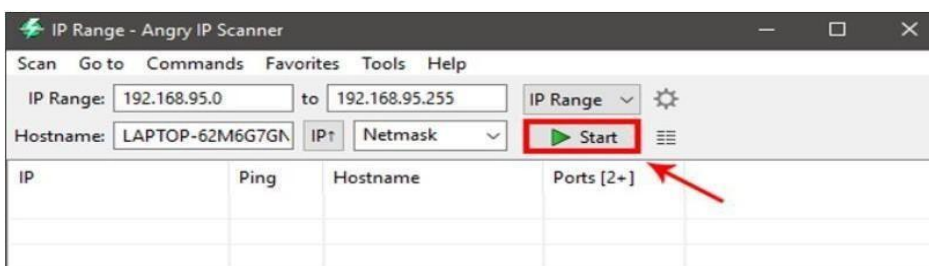
configure the application. To do that, simply navigate to “Tools” and then select the option “Preferences.”



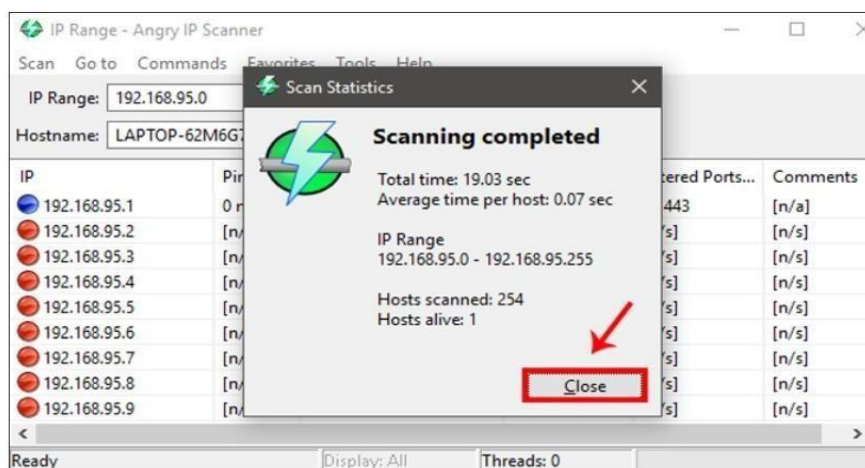
Here, navigate to the “Ports” tab and enter the ports you want to scan in the “Ports Selection” field. As you can see, I’m scanning for port 80 and port 443. If you want to scan a range of ports, then enter the port range like “1-1000.” After you have entered the port number, click OK to save the changes.



Once you are done configuring the Angry IP Scanner, you can continue to scan. To start off, set the scan mode to “IP Range,” enter the IP address range in the “IP address” fields and then click on the button “Start.” For instance, I’ve entered an IP range that is known to have live devices connected to it.



Depending on the number of addresses in the range, it may take some time to complete. Once completed, the application will show you a summary of the scan. The summary includes the number of hosts that are alive and the number of hosts that have open ports. Just click on the button “Close” to continue.



Once you close the summary window, you will see the list of all the IP addresses. You can also see additional details in different “fetcher” columns. In case you are wondering, here’s what the colored dots next to each IP address mean.

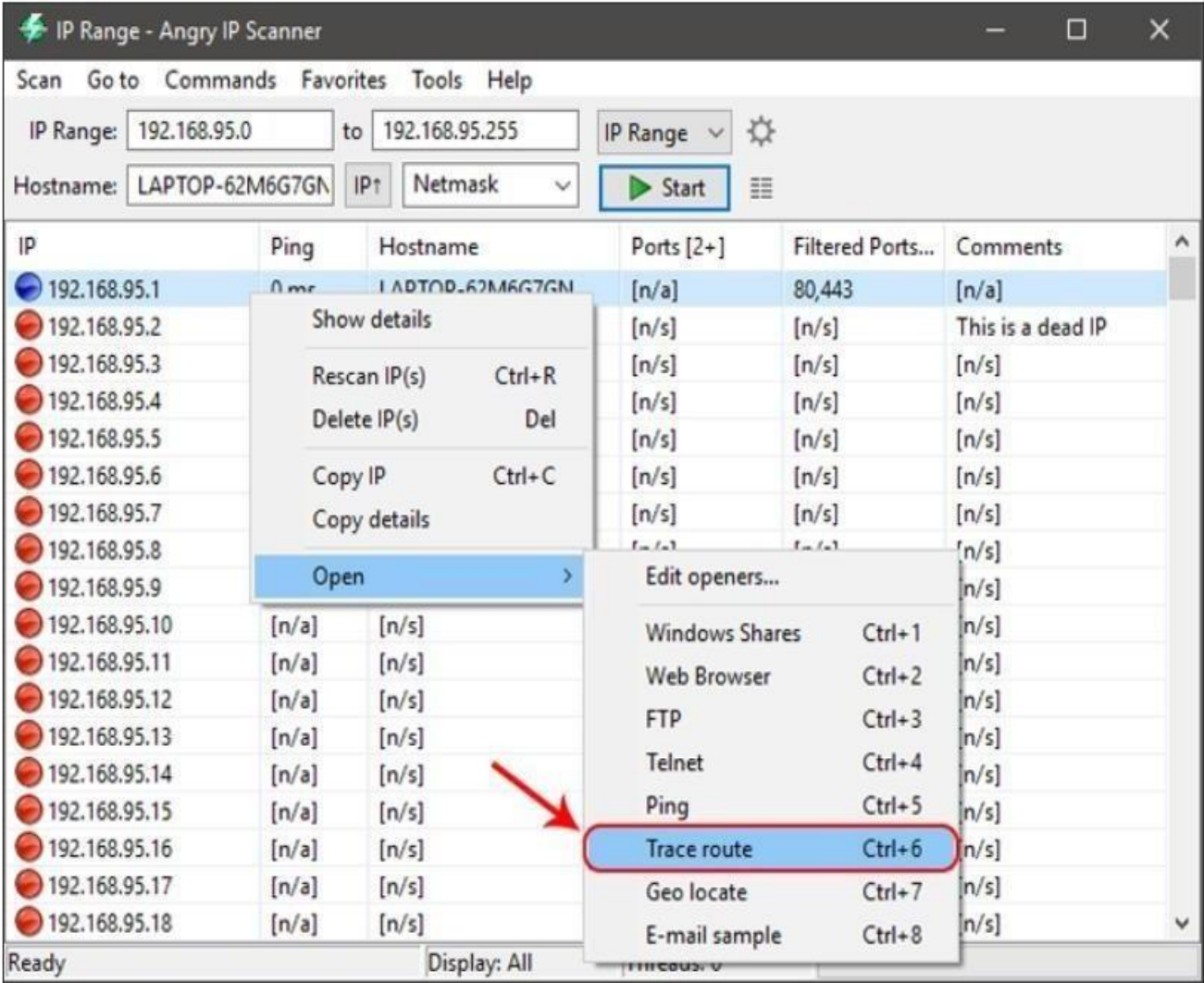
Red: The IP address is inactive, dead or there is no device connected to this IP address.

Blue: The IP address is either active or busy and not responding to the requests sent by Angry IP Scanner. This usually will be your own IP Address.

Green: The IP address is active, and the device connected to it is responding to the requests made by Angry IP Scanner. There may also be open ports.

Apart from copying the details of an IP address, you can also perform a range of different activities on the entries. You can open an IP address in the web browser, do an FTP, trace routing, etc. For instance, if you want to traceroute an IP address, simply right-click on the target IP address. After that, select the option Open and click on

Once you are done scanning an IP address or the IP address range, you can save the scan results. To do that, select the option Scan from the menu bar.



Conclusion: This practical demonstrated how to gather information from LAN-connected PCs using tools like Whois, Nmap, and Angry IP Scanner. We identified active devices, open ports, MAC addresses, and IP details. These techniques are essential for network analysis, ethical hacking, and security auditing. Such network reconnaissance techniques are fundamental in cybersecurity for network mapping, asset identification, and vulnerability assessment.

PRACTICAL: 02

Aim: Experiments with open source firewall/proxy packages like iptables, squid etc.

IP-tables:

- Simply put, iptables is a firewall program for Linux. It will monitor traffic from and to your server using tables. These tables contain sets of rules, called chains, that will filter incoming and outgoing data packets.
- When a packet matches a rule, it is given a target, which can be another chain or one of these special values:
- ACCEPT – will allow the packet to pass through.
- DROP – will not let the packet pass through.
- RETURN – stops the packet from traversing through a chain and tell it to go back to the previous chain.
- PREROUTING - Alters packets **before routing decision**.
- POSTROUTING- Alters packets **after routing decision**, before leaving the system.

Iptables chains:

- In this iptables practical, we are going to work with one of the default tables, called filter. It consists of three chains:
- INPUT – controls incoming packets to the server.
- FORWARD – filters incoming packets that will be forwarded somewhere else. ➤
- OUTPUT – filter packets that are going out from your server.

How to Install and Use Iptables Linux Firewall:

- Installing of Iptables
- Sudo apt-get install iptables

```
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo apt-get install iptables
[sudo] password for jaysd:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libip4tc2 libip6tc2 libnetfilter-conntrack3 libnfnetlink0 libnftables1 libnftnl11 nftables
Suggested packages:
  firewallld
The following NEW packages will be installed:
  iptables libip4tc2 libip6tc2 libnetfilter-conntrack3 libnfnetlink0 libnftables1 libnftnl11 nftables
0 upgraded, 8 newly installed, 0 to remove and 0 not upgraded.
Need to get 983 kB of archives.
After this operation, 4274 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libip4tc2 amd64 1.8.10-3ubuntu2 [23.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libip6tc2 amd64 1.8.10-3ubuntu2 [23.7 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libnfnetlink0 amd64 1.0.2-2build1 [14.8 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libnetfilter-conntrack3 amd64 1.0.9-6build1 [45.2 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libnftnl11 amd64 1.2.6-2build1 [66.0 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 iptables amd64 1.8.10-3ubuntu2 [381 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libnftables1 amd64 1.0.9-1build1 [358 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 nftables amd64 1.0.9-1build1 [69.8 kB]
Fetched 983 kB in 3s (325 kB/s)
Selecting previously unselected package libip4tc2:amd64.
(Reading database ... 40768 files and directories currently installed.)
Preparing to unpack .../0-libip4tc2_1.8.10-3ubuntu2_amd64.deb ...
Unpacking libip4tc2:amd64 (1.8.10-3ubuntu2) ...
Selecting previously unselected package libip6tc2:amd64.
Preparing to unpack .../1-libip6tc2_1.8.10-3ubuntu2_amd64.deb ...
Unpacking libip6tc2:amd64 (1.8.10-3ubuntu2) ...
Selecting previously unselected package libnfnetlink0:amd64.
Preparing to unpack .../2-libnfnetlink0_1.0.2-2build1_amd64.deb ...
Unpacking libnfnetlink0:amd64 (1.0.2-2build1) ...
Selecting previously unselected package libnetfilter-conntrack3:amd64.
Preparing to unpack .../3-libnetfilter-conntrack3_1.0.9-6build1_amd64.deb ...
Unpacking libnetfilter-conntrack3:amd64 (1.0.9-6build1) ...
Selecting previously unselected package libnftnl11:amd64.
Preparing to unpack .../4-libnftnl11_1.2.6-2build1_amd64.deb ...
Unpacking libnftnl11:amd64 (1.2.6-2build1) ...
Selecting previously unselected package iptables.
Preparing to unpack .../5-iptables_1.8.10-3ubuntu2_amd64.deb ...
Unpacking iptables (1.8.10-3ubuntu2) ...
Selecting previously unselected package libnftables1:amd64.
Preparing to unpack .../6-libnftables1_1.0.9-1build1_amd64.deb ...
Unpacking libnftables1:amd64 (1.0.9-1build1) ...
Selecting previously unselected package nftables.
Preparing to unpack .../7-nftables_1.0.9-1build1_amd64.deb ...
Unpacking nftables (1.0.9-1build1) ...
```

- Sudo apt-get update

```
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo apt-get update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1023 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [179 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
```

○ Sudo iptables -L

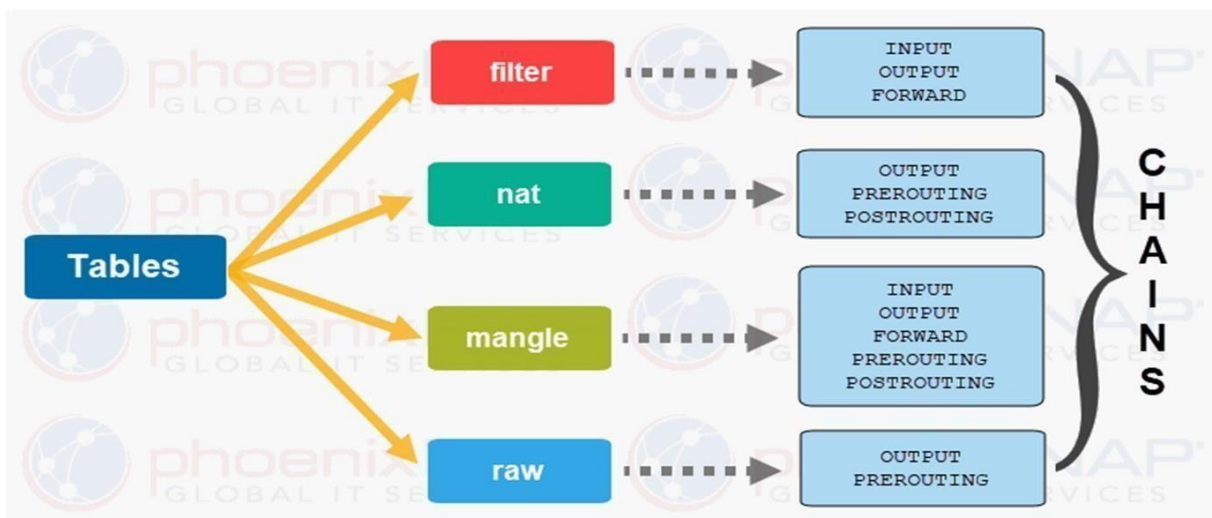
```
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables-L
sudo: iptables-L: command not found
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
jaysd@JAY:/mnt/c/WINDOWS/system32$
```

Types of tables in iptables :

Commands:



- Iptables -A INPUT -s your ip -j DROP
- Iptables -A INPUT -s your ip -j ACCEPT
- To stop https websites
- Iptables -A OUTPUT -p tcp -o eth0 -s your ipaddress --dport 80 -j DROP
- Follow the instructions to create a firewall rule that prevents the communication between two systems via ping.
- Check your ipaddress in both operating system(kali linux and Windows) by follwing commands in the below pictures.


```
jaysd@JAY:/mnt/c/WINDOWS/system32$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.31.62.25 netmask 255.255.240.0 broadcast 172.31.63.255
    inet6 fe80::215:5dff:fe34:beeb prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:34:be:eb txqueuelen 1000 (Ethernet)
    RX packets 27010 bytes 39416967 (39.4 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8994 bytes 680183 (680.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 64 bytes 7422 (7.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 64 bytes 7422 (7.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- Attempt to ping both devices to verify whether they are responsive or not .

```
jaysd@JAY:/mnt/c/WINDOWS/system32$ ping 172.31.62.25
PING 172.31.62.25 (172.31.62.25) 56(84) bytes of data.
64 bytes from 172.31.62.25: icmp_seq=1 ttl=64 time=47.8 ms
64 bytes from 172.31.62.25: icmp_seq=2 ttl=64 time=0.263 ms
64 bytes from 172.31.62.25: icmp_seq=3 ttl=64 time=0.070 ms
64 bytes from 172.31.62.25: icmp_seq=4 ttl=64 time=0.072 ms
64 bytes from 172.31.62.25: icmp_seq=5 ttl=64 time=0.074 ms
```

Please create an iptables rule to stop communication between both devices, as shown in the picture. And try to ping on both devices its working or not.

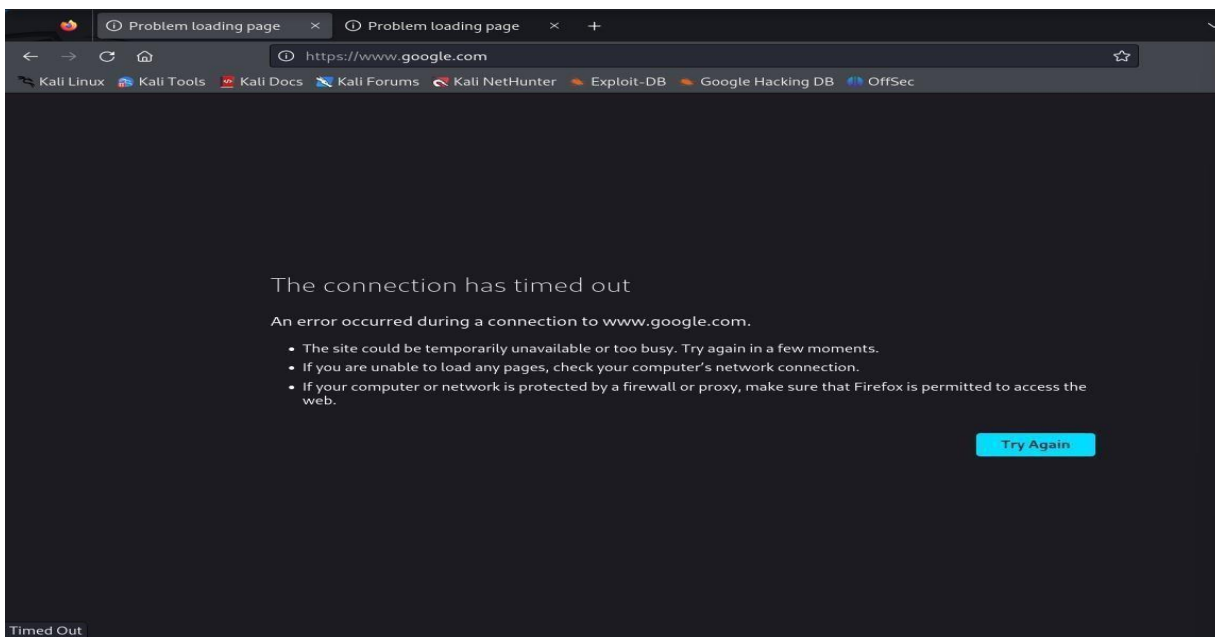
```
rtt min/avg/max/mdev = 0.065/1.627/47.810/8.431 ms
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -A INPUT -s 172.31.62.25 -j DROP
```

- Again if you want resume the communication then you have to write another rule to “ACCEPT” the communication both device by iptable rule. Before you accept the rule We have to remove the previous rule which we have given the “DROP” rule.

```
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -A INPUT -s 172.31.62.25 -j ACCEPT
jaysd@JAY:/mnt/c/WINDOWS/system32$ ping 172.31.62.25
PING 172.31.62.25 (172.31.62.25) 56(84) bytes of data.
```

- To stop https request in our browser by ip address.

```
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -A OUTPUT -p tcp --dport 443 -d 172.31.62.25 -j DROP
jaysd@JAY:/mnt/c/WINDOWS/system32$
```



- To resume the https request in browser by accept rule

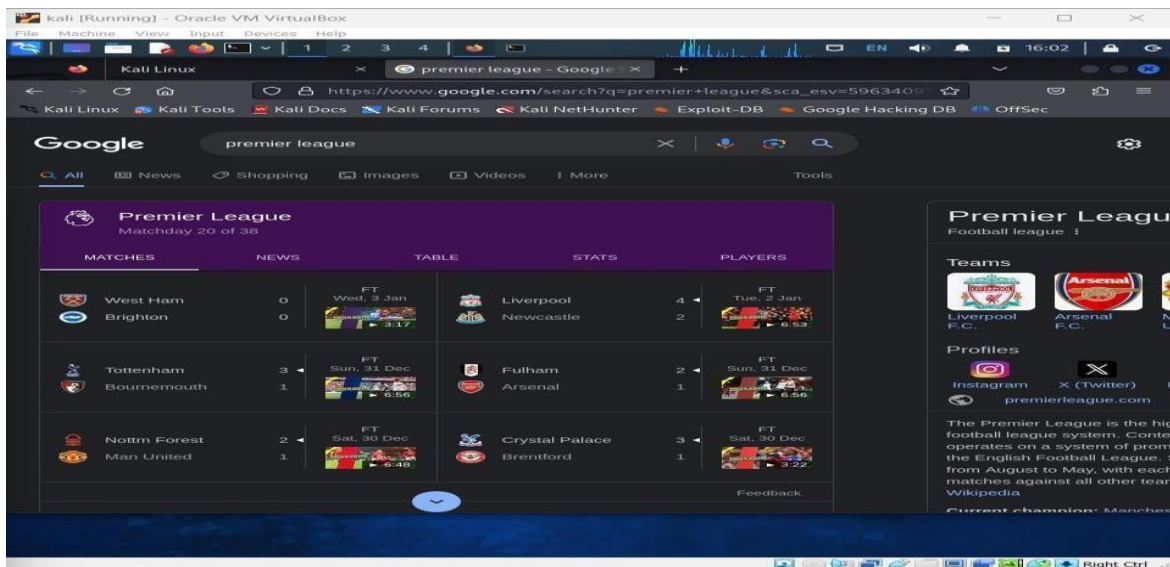
```
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -A OUTPUT -p tcp --dport 443 -d 172.31.62.25 -j ACCEPT
jaysd@JAY:/mnt/c/WINDOWS/system32$
```

```
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -A INPUT -s 172.31.62.25 -j DROP
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -A INPUT -s 172.31.62.25 -j ACCEPT
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -A OUTPUT -p tcp --dport 443 -d 142.250.182.174 -j DROP
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -A OUTPUT -p tcp --dport 443 -d 142.250.182.174 -j ACCEPT
jaysd@JAY:/mnt/c/WINDOWS/system32$ curl https://google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="https://www.google.com/">here</A>.
```

```
jaysd@JAY:/mnt/c/WINDOWS/system32$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 182 packets, 22295 bytes)
pkts bytes target prot opt in out source destination
222 18648 DROP all -- any any 172.31.62.25 anywhere
0 0 DROP all -- any any 172.31.62.25 anywhere
0 0 ACCEPT all -- any any 172.31.62.25 anywhere
0 0 DROP all -- any any 172.31.62.25 anywhere
0 0 ACCEPT all -- any any 172.31.62.25 anywhere
0 0 DROP all -- any any 172.31.62.25 anywhere
0 0 ACCEPT all -- any any 172.31.62.25 anywhere

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination

Chain OUTPUT (policy ACCEPT 98 packets, 7822 bytes)
pkts bytes target prot opt in out source destination
0 0 tcp -- any any anywhere 172.31.62.25 tcp dpt:https
0 0 DROP tcp -- any any anywhere 172.31.62.25 tcp dpt:https
0 0 ACCEPT tcp -- any any anywhere 172.31.62.25 tcp dpt:https
0 0 ACCEPT tcp -- any any anywhere del11s10-in-f14.1e100.net tcp dpt:https
0 0 DROP tcp -- any any anywhere del11s10-in-f14.1e100.net tcp dpt:https
0 0 ACCEPT tcp -- any any anywhere del11s10-in-f14.1e100.net tcp dpt:https
0 0 DROP tcp -- any any anywhere del11s10-in-f14.1e100.net tcp dpt:https
0 0 ACCEPT tcp -- any any anywhere del11s10-in-f14.1e100.net tcp dpt:https
jaysd@JAY:/mnt/c/WINDOWS/system32$
```



Conclusion: ➤ In this practical, we learned how to use iptables to monitor incoming and outgoing traffic on your server through tables. Additionally, we explored the mechanisms for dropping and accepting services when communicating with devices.

PRACTICAL: 03

Aim: Implementation of Steganography.

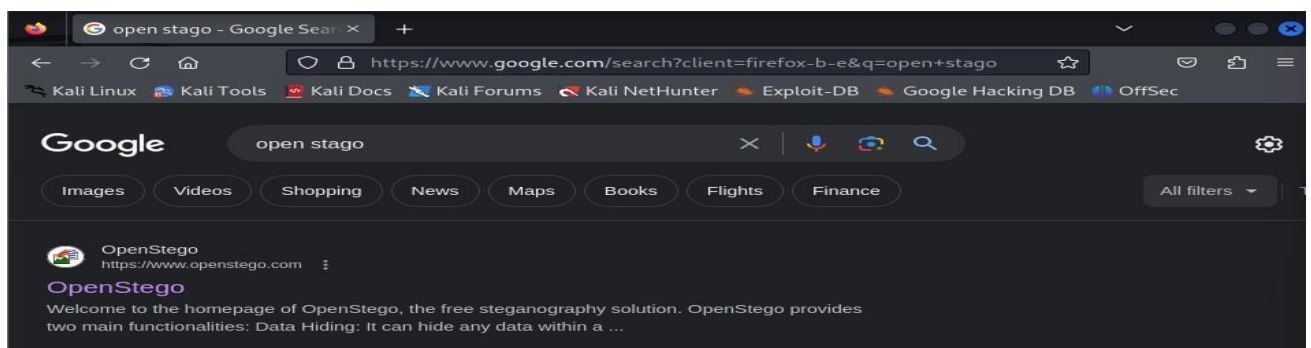
Theory:

Steganography:

- The root “steganos” is Greek for “hidden” or “covered,” and the root “graph” is Greek for “to write.” Steganography is the practice of hiding a secret message inside of (or even on top of) something that is not secret.
- Examples of steganography involve embedding a secret piece of text inside of a picture. Or hiding a secret message or script inside of a Word or Excel document. The purpose of steganography is to conceal and deceive. It is a form of covert communication and can involve the use of any medium to hide messages.
- It's not a form of cryptography, because it doesn't involve scrambling data or using a key. Instead, it is a form of data hiding and can be executed in clever way
- Types of Steganography:
 1. Text Steganography
 2. Image Steganography
 3. Video Steganography
 4. Audio Steganography
 5. Network Steganography

Steps To follow:

Step 1: Open kali and go to the browser, search for “open stego” tool shon in below figure.



Step 2: Go to open stego site and download it from their.

Step 3: To install the tool open terminal from the download directory.

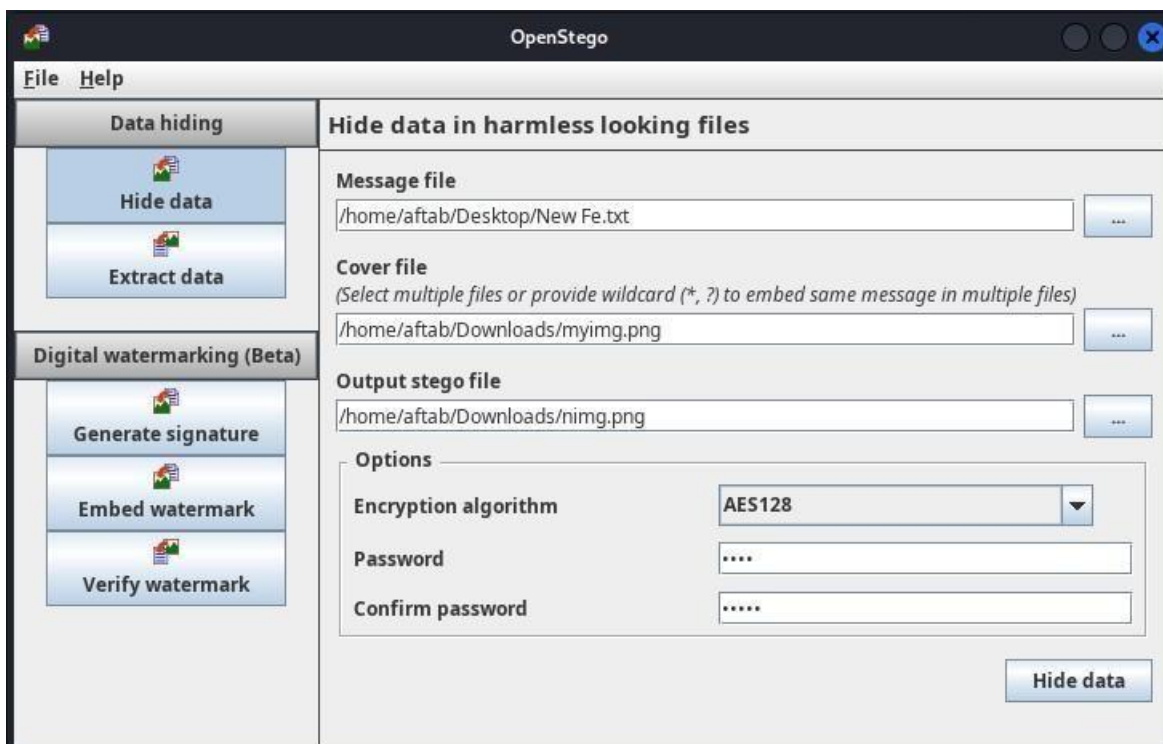
Step 4: Then run command “sudo dpkg -i openstego_0.8.6-1_all.deb” as depicted in below figure.

Step 5: Create the text file of the message you want to embed.

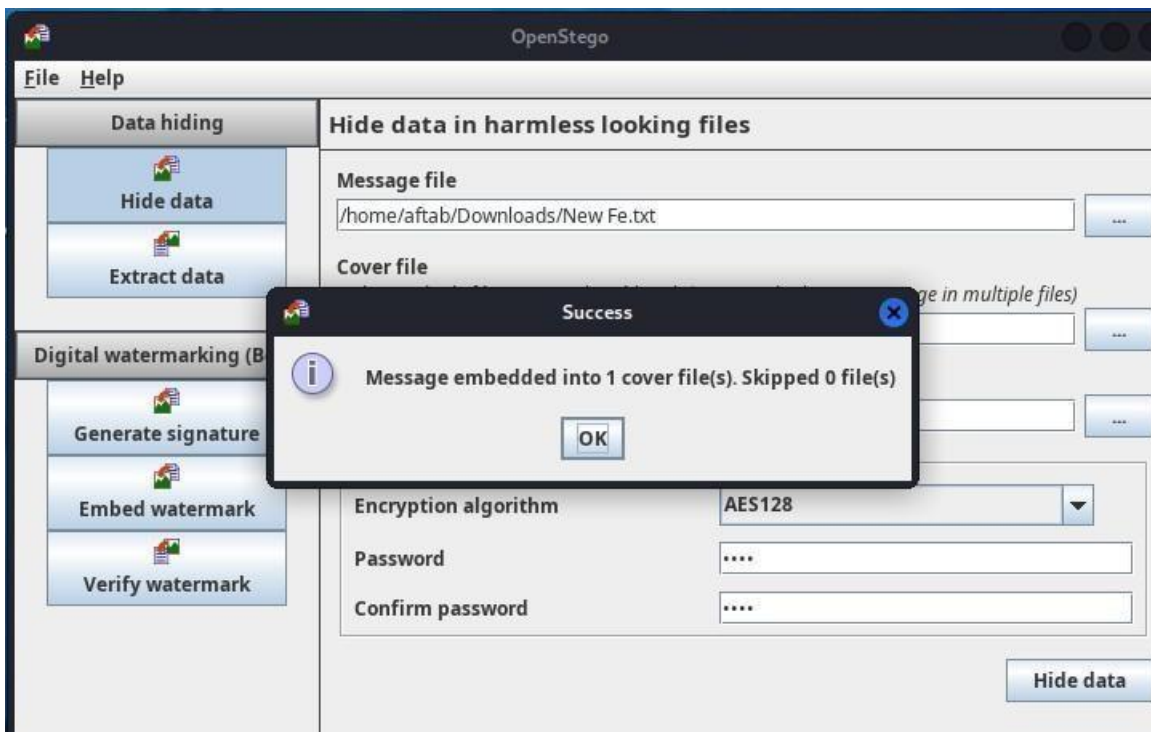


Step 6: After the successful creation of text file save it and launch open stego application.

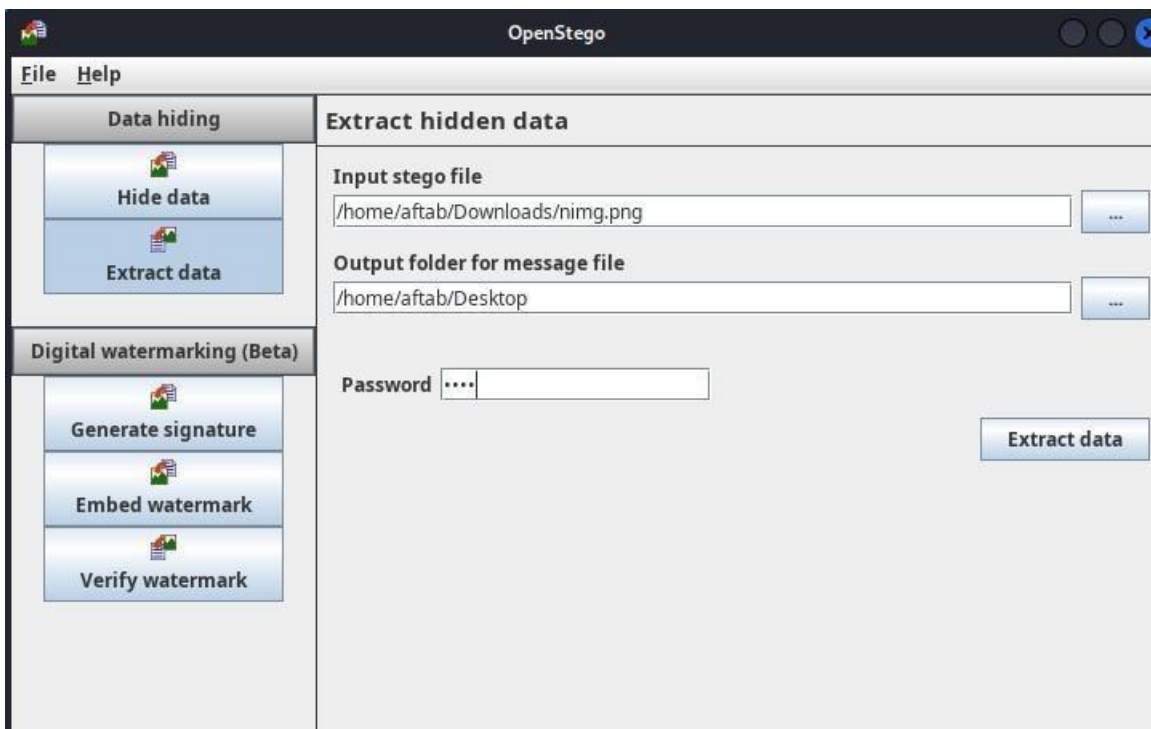
Step 7: Give all the require text file path and image file in which the message is goin to be embedded create a password for encryption.



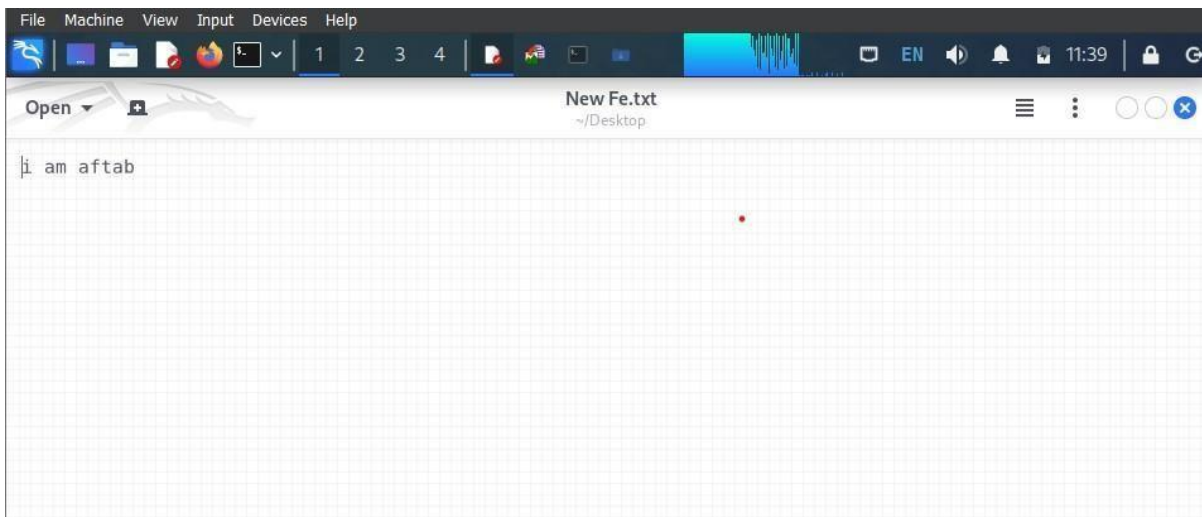
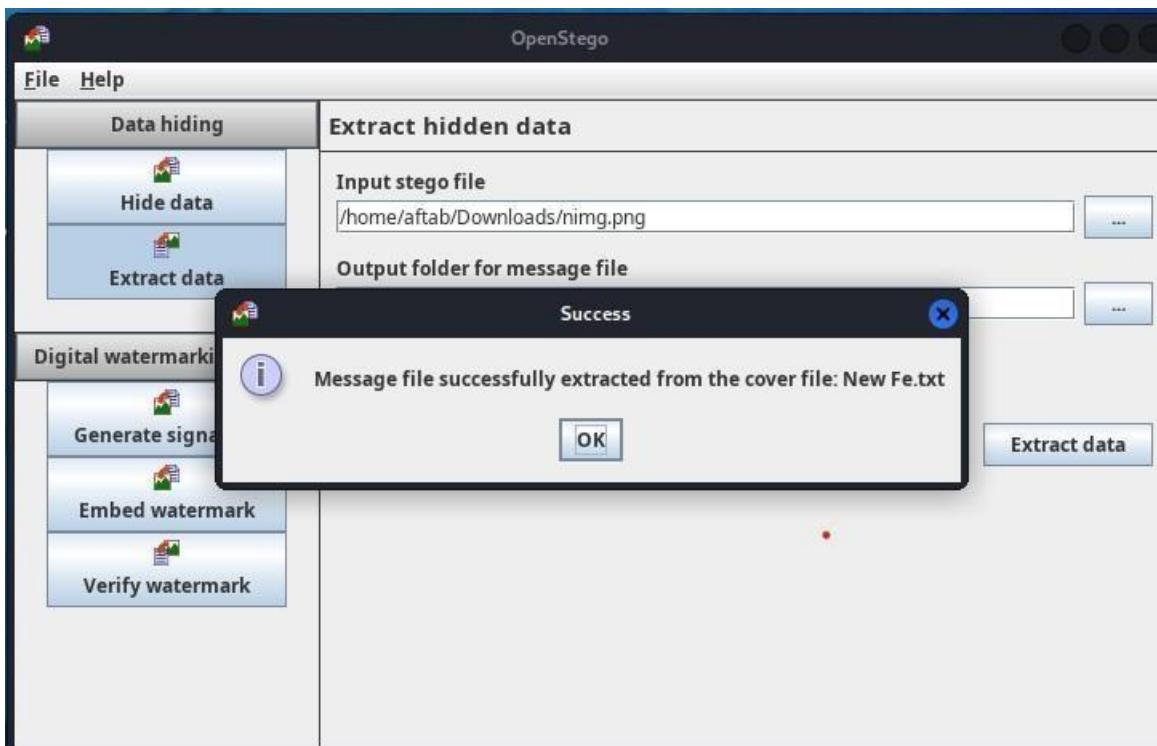
Step 8: after filling all details click on hide to perform data hiding task.



Step 9: To check the message embedded in image go to open stego application.



Step 10: By giving the necessary files path and password the data will be extracted.



Conclusion:

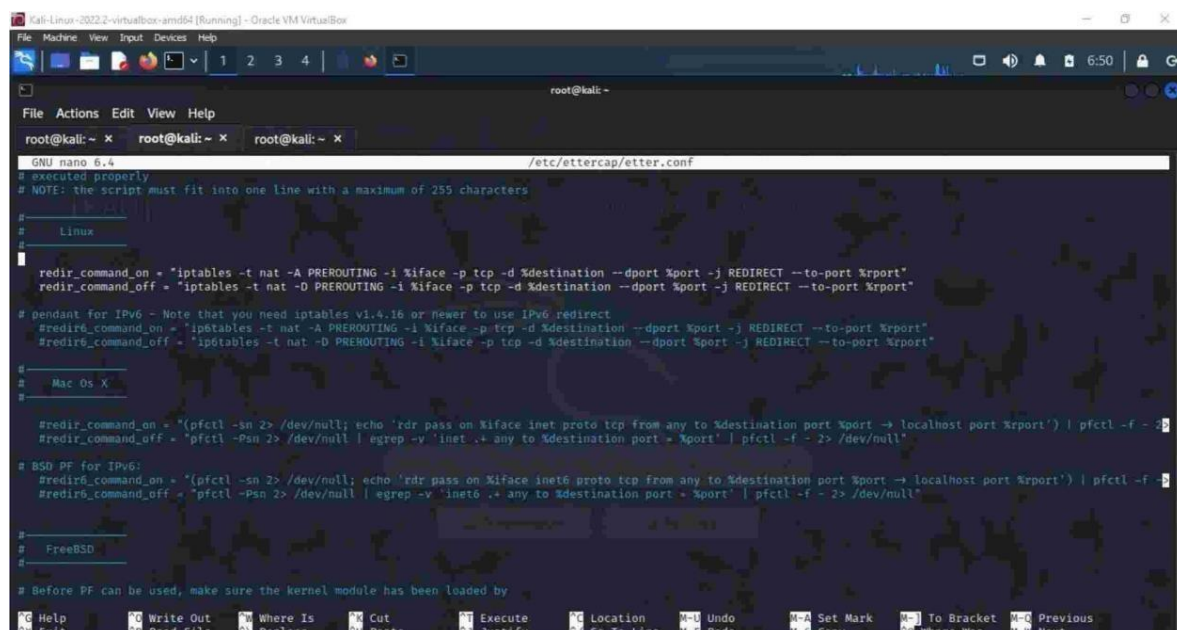
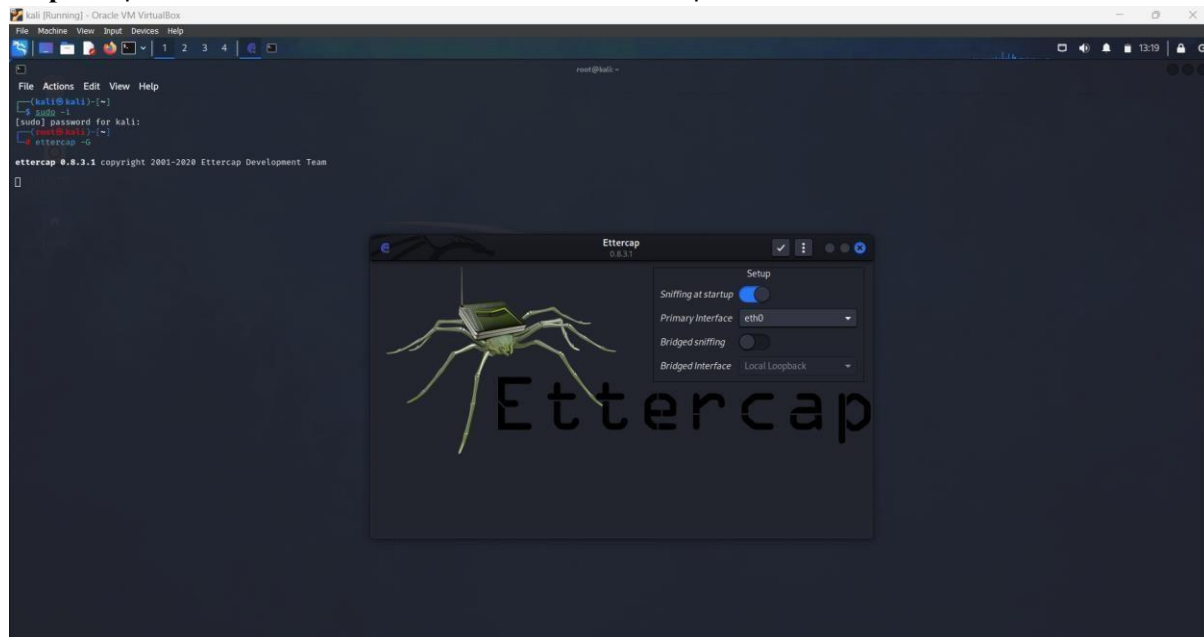
In this practical we learned how to hide a message with the image and encrypt it , also how to extract the hidden message from the image.

PRACTICAL: 04

Aim: Implementation of MITM- attack using Wireshark/network sniffers.

Requirements: Linux, Ettercap, Advance IP scanner, Target IP.

Step-1: Open terminal in kali and write command “Ettercap -G”.




```

GNU nano 6.4 /etc/ettercap/etter.dns
# domain13.com MX XXXX:XXXX:YY
#
# or for WINS query:
# workgroup WINS 127.0.0.1 [TTL]
# PC* WINS 127.0.0.1
#
# or for SRV query (either IPv4 or IPv6):
# service_tcp_udp.domain SRV 192.168.1.10:port [TTL]
# service_tcp_udp.domain SRV [2001:db8::3]:port
#
# or for TXT query (value must be wrapped in double quotes):
# google.com TXT "v-spfl ip4:192.168.0.3/32 -all" [TTL]
#
# NOTE: the wildcarded hosts can't be used to poison the PTR requests
# so if you want to reverse poison you have to specify a plain
# host. (look at the www.microsoft.com example)
#
# NOTE: Default DNS TTL is 3600s (1 hour). All TTL fields are optional.
#
# NOTE: IPv6 specific do not work because ettercap has been built without
# IPv6 support. Therefore the IPv6 specific examples has been
# commented out to avoid ettercap throwing warnings during startup.
#
# vim:ts=8:nowrap:
facebook.com A 192.168.182.10
*.facebook.com A 192.168.182.10
www.facebook.com PTR 192.168.182.10
  
```

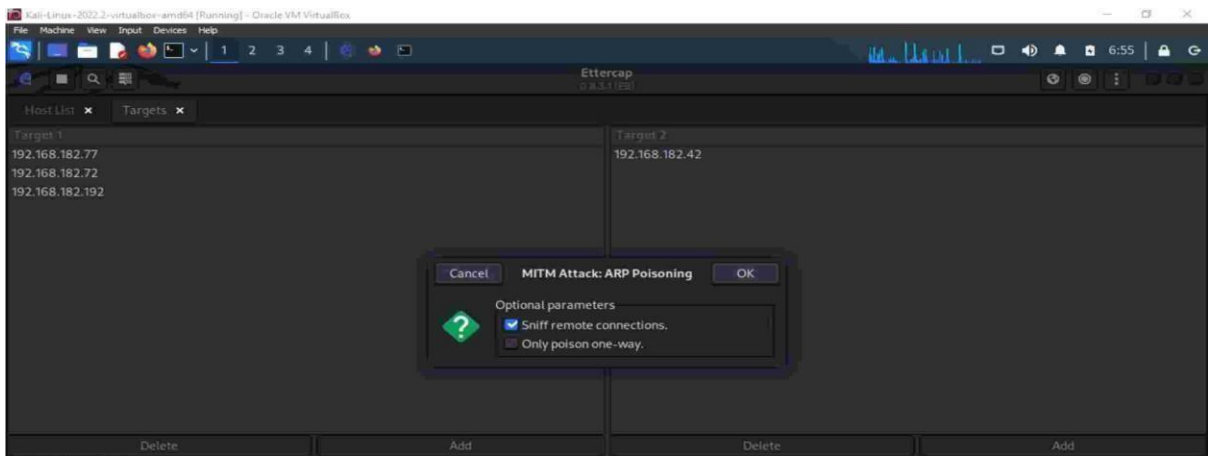
Step-2: Open hosts list.

IP Address	MAC Address	Description
192.168.182.42	72:6F:E5:58:8E:51	
192.168.182.72	00:45:E2:5D:49:7B	
192.168.182.77	80:91:33:93:8F:E7	
192.168.182.192	80:91:33:93:8F:E7	

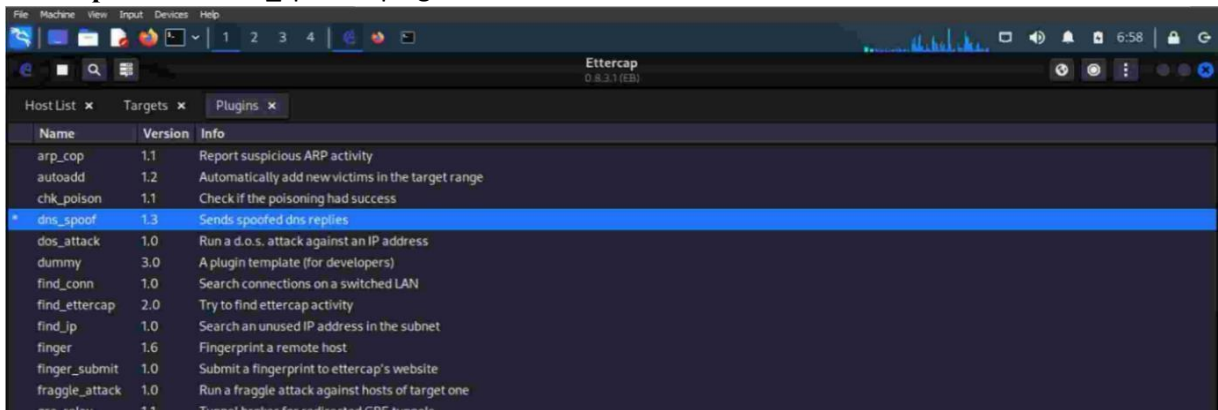
Delete Host Add to Target 1 Add to Target 2

Lua: no scripts were specified, not starting up!
 Randomizing 255 hosts for scanning...
 Scanning the whole netmask for 255 hosts...
 4 hosts added to the hosts list...
 Starting Unified sniffing...

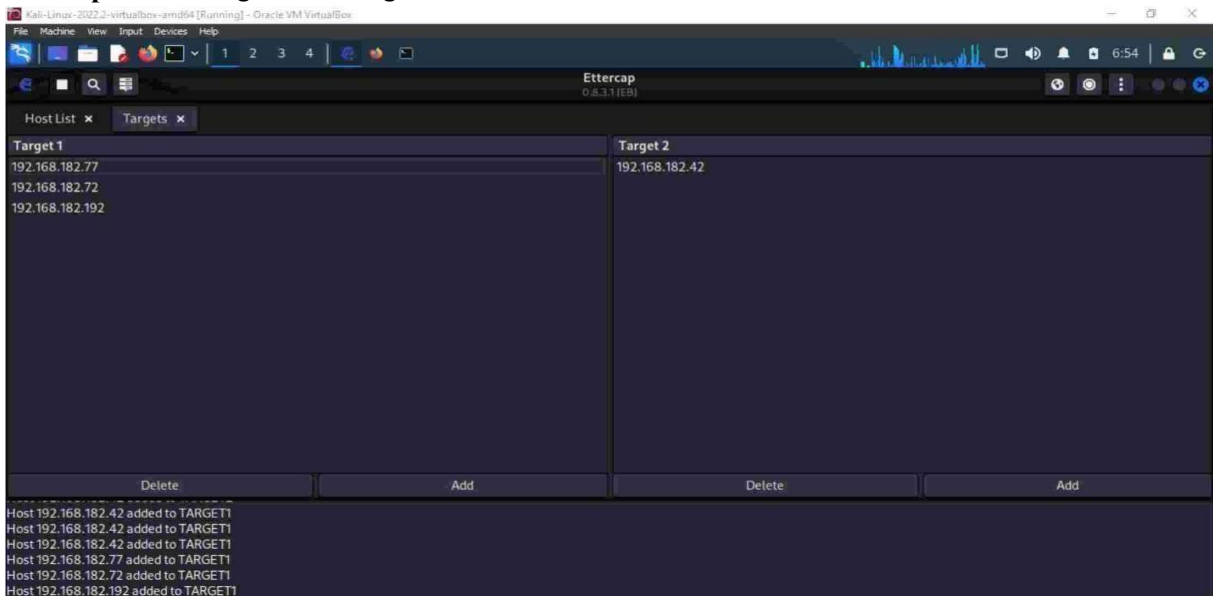
Step 3: Select sniff remote connections in MITM attack: ARP Poisoning.



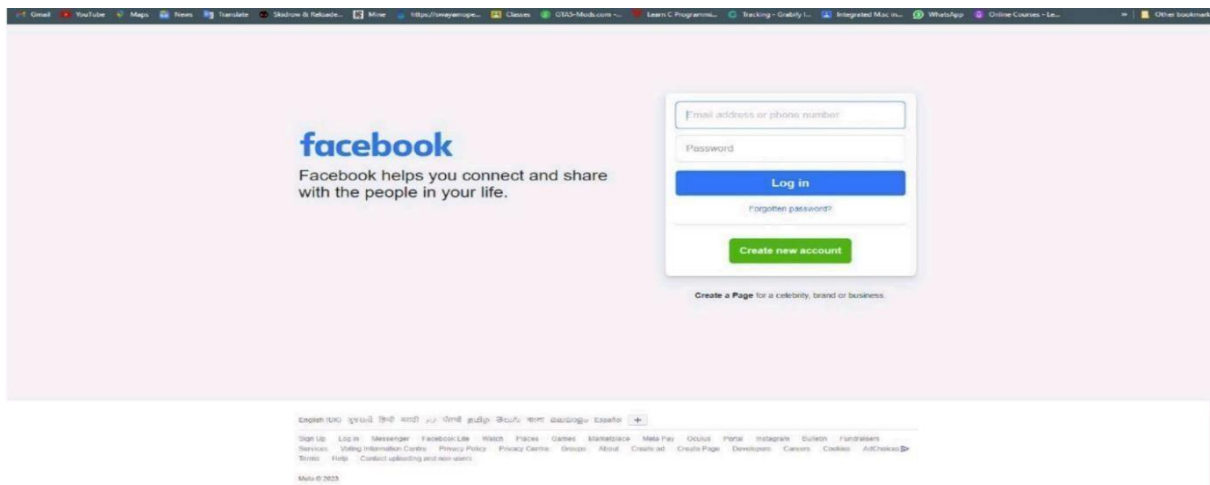
Step-4: Select dns_spoof in plugins.



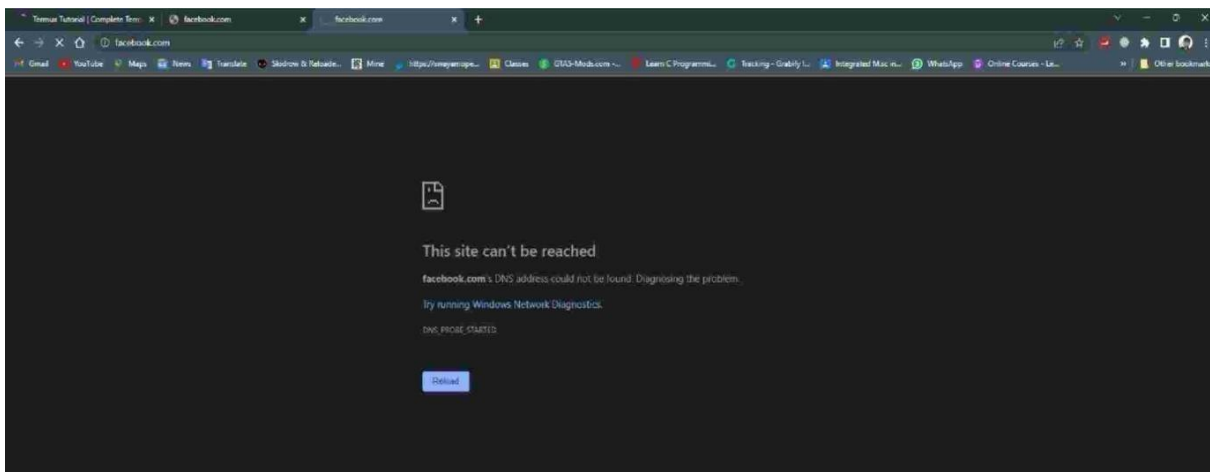
Step-5: Add Target1 and Target2.



- Before



• After



Conclusion: In this practical, we learn about ARP spoofing with help of Ettercap or tcp dump. It is often performed for security testing purposes with proper authorization. Unauthorized ARP spoofing can be illegal and unethical, leading to severe consequences.